

CS401: Database Management Systems

Unit 1: Transaction Processing & ACID Properties

A transaction is a logical unit of database processing that includes one or more database operations.

To ensure integrity, transactions must adhere to the ACID properties:

1. Atomicity: Requires that all operations of a transaction are executed completely, or none are (all-or-nothing).
2. Consistency: Ensures that a transaction brings the database from one valid state to another valid state.
3. Isolation: Ensures that concurrent execution of transactions results in a state that would be obtained if transactions were executed sequentially.
4. Durability: Guarantees that once a transaction commits, its results are permanently saved in non-volatile storage even in the event of a system failure.